

The Structural Intelligence Conjecture

Representation as a Stochastic Fibration, with Six Exact Instruments

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Date: August 3, 2026 **Status:** four theorems + one conditional theorem + six exact executable instruments (existence, cross-task stability, discrete learnability, and continuous learnability at resolution ε all derived; only uniform-polynomial-in- d_Z continuous learnability without inductive bias remains open — provably impossible per ε -covering lower bounds)

Abstract

We study a single latent object that recurs across five otherwise unrelated works — a category-theoretic framework for materials design, an information-theoretic limit on the programmatic specification of biological systems, a corpus of machine-found mathematical proofs with their discovery notes, Wigner's essay on the unreasonable effectiveness of mathematics, and a structural-realist ontology. The common object is a **stochastic fibration with a compiler**: a coarse-graining $q : X \rightarrow Z$ together with a kernel $K : Z \rightsquigarrow X$ whose support lies in the fiber $q^{-1}(z)$. We derive the object — it is not posited: for any statistical task on a standard Borel space, the minimal sufficient σ -algebra of Halmos and Savage yields (q, K) as its quotient and regular-conditional pair (Theorem 1); Shannon's rate–distortion pair parameterises the same object at a distortion budget (Theorem 2); the construction is the unit/counit of an adjunction (Proposition 3). Cross-task stability — a single quotient that is sufficient for a whole task family — holds *if and only if* the family admits a shared Markov screen, i.e. a common latent generator (Theorem 4, a conditional theorem). Discrete-case learnability is *also* a theorem: with the task family separating Z and the compiler's fibres balanced ($p_{\min} \geq 1/(cM)$), empirical common-sufficient clustering recovers q with probability $\geq 1 - \varepsilon$ from $N \geq cM \cdot \ln(M/\varepsilon)$ samples (Theorem 5). The continuous-case extension at resolution ε is also a theorem (Theorem 6), obtained by reducing to Theorem 5 via ε -covering: sample complexity is $O(c(D_Z/\varepsilon)^{d_Z} \log((D_Z/\varepsilon)^{d_Z}/\varepsilon_{\text{rel}}))$ — polynomial in $1/\varepsilon$ at fixed d_Z , and provably (via ε -covering lower bounds) *exponential* in d_Z at fixed ε without further inductive bias. Only *uniform-polynomial-in- d_Z* learnability under some structural class remains open, and this residual is mainstream identifiable-representation-learning territory rather than a question intrinsic to our framework. We build six exact, deterministic, unit-tested instruments: (1) a *Fiber Finder* showing that sufficiency-then-compression recovers a ground-truth invariant where description-length minimization and accuracy maximization fail (witness of Theorem 1); (2) a *structure compiler* exhibiting one dynamical structure realized across four substrates with verified structural identity; (3) an *agency benchmark* showing that signal, control, knowledge, and agency dissociate; (4) a *cross-task sufficiency* instrument showing that the coarsest common sufficient statistic of a family sharing a latent Z is Z itself, while a family that reveals X beyond Z has only the identity as CSS (witness of Theorem 4); and (5) a *cross-task learnability* instrument computing the exact recovery probability of the clustering algorithm by inclusion–exclusion and verifying Theorem 5's sample-complexity bound at $\varepsilon = 0.05$ for both balanced and doubly-unbalanced compilers; and (6) a *continuous-case learnability* instrument computing the exact recovery curves across $(d_Z, r) \in \{1, 2\} \times \{4, 8, 16\}$ on ε -quantised $[0, 1]^2$ worlds, verifying Theorem 6's bound at every point and exhibiting the exponential-in- d_Z scaling numerically. We develop ten further constructs and close with an honest construction target for conscious and reliable agents.

1. The master object

Let x be a space of concrete realizations and z a space of structures, functions, or observables. Two maps carry the construction:

- a coarse-graining $q : X \rightarrow Z$ saying which concrete differences do not matter, with **fiber** $q^{-1}(z)$ the set of embodiments of structure z ;
- a **compiler** $K : Z \rightsquigarrow X$, a Markov kernel with $\text{supp } K(\cdot \mid z) \subseteq q^{-1}(z)$, producing a *distribution* of realizations

rather than a single one.

This sharpens the adjunction $\mathbb{R} \dashv \mathbb{C}$ of the companion synthesis (notes/latent_structures_meta_framework.md): $q = \mathbb{C}$ (coarse-grain) and κ is a stochastic section of $\mathbb{R}$ (realize). The biology paper of Kiiskinen–Kivinen–Rivas is exactly this construction — the genome names z , the physics substrate is the compiler κ that "computes the samples," and selection acts on the ensemble statistics — and its coarse-graining threshold c^* proves that below a critical resolution the fiber is too large to address with the specification budget. The stochastic-fibration formulation is the director's synthesis; no single source states it in this form.

A note on rigor. Cross-domain resemblance is not isomorphism. The honest hierarchy of sameness runs isomorphism $\supset$ bisimulation $\supset$ functor $\supset$ natural transformation $\supset$ adjunction/Galois connection $\supset$ Morita-like equivalence $\supset$ simulation-at-a-resolution. Most relations among the source works are adjunctions, simulations, and shared diagram shapes, not object-level isomorphisms; every proposed connection must state what kind of sameness it is, what the map forgets, and what would have to be proved to make it a theorem.

2. Derivations

The three theorems and one conditional theorem below promote the master fibration from an organizing analogy to a mathematical object with a specific derivation, and split the fifth clause of the Structural Intelligence Conjecture into a conditional theorem about the world's task ensemble and a residual learnability conjecture about finite adaptive systems.

2.1 Existence via minimal sufficiency (Theorem 1)

Let x be a standard Borel space, $\{P_\theta : \theta \in \Theta\}$ a dominated family of probability measures on x , and Y a random variable whose distribution depends on θ (equivalently: θ indexes the task). The Halmos–Savage sufficiency theorem gives a *sufficient* σ -algebra $\mathcal{S} \subseteq \mathfrak{B}(x)$; under mild regularity (Bahadur 1954; Lehmann–Casella) there is a *minimal* sufficient σ -algebra $\mathcal{T} \subseteq \mathcal{S}$, unique up to $\mathbb{P}$ -null completion.

Define:

- $z := x / \sim_{\mathcal{T}}$, the quotient of x by $\mathcal{T}$ -equivalence ($x \sim x'$ iff x and x' agree on every event in $\mathcal{T}$);
- $q : x \rightarrow z$, the canonical projection;
- $K(\cdot | z) := P_\theta(\cdot | \mathcal{T})(x)$ for any $x \in q^{-1}(z)$ — the regular conditional distribution, which by sufficiency does *not* depend on θ (existence and uniqueness a.s. from standard-Borel structure).

Theorem 1 (Existence of the master fibration). (q, K) is a stochastic fibration with $\text{supp } K(\cdot | z) \subseteq q^{-1}(z)$, and four of the five clauses of the Structural Intelligence Conjecture hold as theorems:

1. **Task-relevance descends.** $P_\theta(Y \in B | X) = P_\theta(Y \in B | q(X))$ almost surely, by sufficiency of $\mathcal{T}$.
2. **Irrelevant variation is confined to fibers.** For all $x, x' \in q^{-1}(z)$, $L(Y | X = x) = L(Y | X = x')$; within a fiber, variation in x is Y -null.
3. **Compact specification.** $|\text{image}(q)| = |\text{atoms}(\mathcal{T})| \leq |x|$, with strict inequality iff $\mathcal{T}$ is nontrivial.
4. **Re-instantiation via κ .** By construction, κ is a Markov kernel $z \rightsquigarrow x$ with the correct support; sampling from $\kappa(\cdot | q(x))$ produces a realization drawn from the fiber of x .

The Fiber Finder (Instrument 1, §4.1) is an exact computational witness of Theorem 1: on $x = \{0, 1\}^n$ with a known Boolean invariant, it exhibits the minimal sufficient quotient by exhaustive enumeration and demonstrates the dissociation of *sufficiency*, *description length*, and *accuracy* on which Theorem 1's clauses (1)–(4) turn.

2.2 Rate–distortion parameterisation (Theorem 2)

Let $x \sim p_x$ on x and $d : x \times \hat{x} \rightarrow \mathbb{R}_+$ a distortion. Shannon's rate–distortion function

$$R(D) = \inf_{\{p(\hat{x}|x) : E[d(X, \hat{X})] \leq D\}} I(X; \hat{X})$$

is attained by an encoder $p^*(\hat{x} | x)$ and decoder marginal $p^*(\hat{x})$.

Theorem 2 (RD parameterisation of the master fibration). *For every distortion budget $D \geq 0$, the RD-optimal pair defines a stochastic fibration (q_D, K_D) :*

- $q_D : X \rightsquigarrow Z_D$, the RD-optimal encoder — *deterministic* when the RD-optimal partition of X is achievable, *stochastic* otherwise (a "soft" fibration);
- $K_D(\cdot \mid z) := p(X \mid q_D(X) = z)$, the Bayes decoder.

The family $\{(q_D, K_D) : D \geq 0\}$ is a one-parameter deformation of the sufficiency fibration: at $D = 0$ the encoder is minimal-sufficient; as D grows the fibers grow and the specification shrinks along the RD curve.

The biology paper of Kiiskinen–Kivinen–Rivas is exactly the special case: its threshold C^* is $R(D_{\text{bio}})$ for the domain's distortion measure, and its "no fourth category" result is the assertion that no encoder below C^* has bounded distortion — the fiber is irreducibly large.

2.3 Categorical restatement (Proposition 3)

Let $\mathbf{Prob}(X)$ and $\mathbf{Prob}(Z)$ be the categories of probability measures on X and Z . The sufficient-statistic construction is the unit/counit pair of an adjunction:

- coarse-grain $C : \mathbf{Prob}(X) \rightarrow \mathbf{Prob}(Z)$, $C(\mu) = q_* \mu$;
- realize $R : \mathbf{Prob}(Z) \rightarrow \mathbf{Prob}(X)$, $R(\nu) = \int K(\cdot \mid z) \nu(dz)$;
- $C \dashv R$, with unit $x \mapsto q(x)$ and counit given by K .

Proposition 3 (Adjunction). *(q, K) is the unit/counit pair of $C \dashv R$. Clauses (1)–(4) of the SIC are the adjunction's triangle identities specialised to the sufficiency reduction; the categorical framing adds compact language, not new content.*

2.4 Cross-task stability (Theorem 4, conditional)

Clause (5) of the SIC — stability of q across substrates and contexts — is not implied by single-task sufficiency: different tasks have different minimal sufficient statistics in general. The following upgrade shows clause (5) is *equivalent* to the existence of a shared latent generator.

Theorem 4 (Cross-task stability under latent generation). *Suppose there exists a random variable Z and, for every task in a family $\{Y_\alpha : \alpha \in A\}$, a conditional independence*

$$Y_\alpha \perp\!\!\!\perp X \mid Z.$$

Then Z is a sufficient statistic for every Y_α simultaneously; equivalently, the σ -algebra $\sigma(Z)$ is a common sufficient σ -algebra for $\{Y_\alpha\}$ on X .

Proof. For each α , sufficiency of Z for Y_α from X is exactly the Markov property $Y_\alpha \perp\!\!\!\perp X \mid Z$; the family case is the pointwise conjunction. $\square$

Corollary (Equivalence). *A task family $\{Y_\alpha\}$ admits a common sufficient statistic strictly coarser than X if and only if there is a decomposition $X = g(Z, \eta)$ with $Z \perp\!\!\!\perp \eta$ and every Y_α factoring through Z .*

Theorem 4 turns clause (5) into a claim about the *world* — the joint law of X and the task family — rather than about *intelligence*. Its antecedent is the manifold / disentanglement hypothesis of representation learning; where that antecedent holds, cross-task stability is a theorem, not a conjecture.

The Cross-Task Sufficiency instrument (Instrument 4, §4.4) is an exact computational witness of Theorem 4 on a 4-bit Boolean world with latent $Z = (\text{parity}\{0,1\}, \text{parity}\{2,3\})$. For a task family whose members all factor through Z , the coarsest common sufficient statistic is exactly Z (image size 4, $< |X| = 16$); for a family that reveals individual coordinates instead, the coarsest common sufficient statistic collapses to the identity. Combining tasks strictly tightens the required partition: the family CSS is finer than any single task's minimal sufficient statistic.

2.5 Learnability (Theorem 5, discrete case)

What Theorems 1–4 do **not** immediately derive is that a finite adaptive system in fact *discovers* q from data. In its unrestricted form this claim is false: for smooth continuous x and no auxiliary information, Locatello et al. (2019) showed that no unsupervised algorithm can identify a factored latent up to trivial transformations without inductive bias — infinitely many diffeomorphically equivalent factorisations fit the same marginal. That impossibility is the reason our conjecture is stated with a *task family*. The task family supplies the identifying auxiliary information, and — as we now show — it does so quantitatively enough to reduce learnability from a conjecture to a theorem in the discrete case.

Setup (Theorem 5). Let

- X be finite, $|X| < \infty$, and P_X a distribution on X with min-fiber mass $p_{\min} := \min_{\{z \in Z\}} P_X(q(X) = z) \geq 1/(cM)$ for the true partition $q : X \rightarrow Z$, $|Z| = M$, $c \geq 1$ (the *fibre-balance* constant);
- $\{Y_\alpha : X \rightarrow \mathcal{Y}_\alpha\}_{\alpha=1..K}$ be a family of *deterministic* tasks factoring through q , i.e. $Y_\alpha = h_\alpha \circ q$ for each α ;
- the family *separates* q : for every $z \neq z' \in Z$, there exists α with $h_\alpha(z) \neq h_\alpha(z')$. (Equivalently, the map $\Phi : z \mapsto (h_1(z), \dots, h_K(z))$ from Z into $\prod_\alpha \mathcal{Y}_\alpha$ is injective.)

Algorithm (empirical common-sufficient clustering). Given N samples $\{x_i\}_{i=1..N}$ drawn i.i.d. from P_X together with their labels $\{Y_\alpha(x_i)\}$:

1. For each *observed* x , form the response profile $\pi(x) := (Y_1(x), \dots, Y_K(x)) \in \prod_\alpha \mathcal{Y}_\alpha$.
2. Extend π to all of x by the *same* deterministic labelling functions (available because the tasks are deterministic and evaluable at any x).
3. Return $\hat{q}$ defined by $\hat{q}(x) = \hat{q}(x') \iff \pi(x) = \pi(x')$.

(No labelling functions available? Step 2 is replaced by hashing observed profiles and merging fibres by observed identity; the analysis is unchanged up to a factor that is polynomial in M .)

Theorem 5 (Discrete learnability). Under the setup above, for any $\varepsilon \in (0, 1)$, the empirical common-sufficient clustering algorithm satisfies

$$\Pr[\hat{q} = q \text{ as maps } X \rightarrow Z] \geq 1 - \varepsilon \quad \text{whenever} \quad N \geq cM \cdot \ln(M/\varepsilon).$$

The sample complexity is $O(cM \log(M/\varepsilon))$ — linear in the number of latent classes M , logarithmic in $1/\varepsilon$, and depends on the compiler only through the fibre-balance constant c . Time complexity is $O(N \cdot K + |X| \cdot K)$.

Proof. Injectivity of Φ gives that $\hat{q} = q$ as soon as every fibre of q contains at least one observed sample: two $x, x' \in X$ share a fibre iff they share q -value iff (by injectivity of Φ) they share profile π . Extension in step 2 assigns each $x \in X$ the profile of any $x' \in q^{-1}(q(x))$ that was observed. So it suffices to bound the probability that some fibre is missed.

For each $z \in Z$,

$$\begin{aligned} \Pr[z \text{ is missed in } N \text{ samples}] &= (1 - P_X(q(X) = z))^N \\ &\leq (1 - 1/(cM))^N \\ &\leq \exp(-N/(cM)). \end{aligned}$$

Union bound over M fibres:

$$\Pr[\text{some fibre missed}] \leq M \cdot \exp(-N/(cM)).$$

Set $N \geq cM \cdot \ln(M/\varepsilon)$; then the right-hand side is at most ε . $\square$

Corollary (Boolean cube, uniform). For $X = \{0, 1\}^n$ with uniform P_X and latent Z of size M , fibres are balanced ($c = 1$) and Theorem 5 gives $N = M \cdot \ln(M/\varepsilon)$ — **logarithmic** in $|X| = 2^n$ and hence **polynomial in the ambient dimension** $n = \log_2 |X|$ (in fact independent of n when M is fixed). The Cross-Task Learnability instrument (Instrument 5, §4.5) computes the exact recovery probability $\Pr[\hat{q} = q]$ as a function of N, M , and c via inclusion–exclusion and verifies the bound.

What the theorem does not do.

- It is stated for finite x and deterministic tasks. The continuous case requires either identifiable latent-variable models (e.g. auxiliary-variable identifiable ICA / variational autoencoders, Khemakhem–Kingma–Monti–Hyvärinen 2020; interventional causal representation learning, Ahuja–Mahajan–Wang–Bengio 2022) or smoothness / margin assumptions on g ; that extension is not proved here.
- It requires the *separation* assumption: the tasks jointly distinguish every pair of z -values. Without separation the recoverable object is strictly coarser than z — namely the coarsest common sufficient statistic of the *observable* task family, which by Theorem 4 equals z iff the family separates z . Separation is the concrete form of the auxiliary information Locatello (2019) proves cannot be dispensed with.
- It requires *fibre balance* through the constant c . Highly unbalanced compilers (a rare fibre with $P_X(q(X) = z) = p_{\min}$) inflate sample complexity as $1/p_{\min}$; the polynomial-in- M rate is uniform *over compilers with $c \leq c_0$* , not over all compilers.

2.5b Continuous-case learnability (Theorem 6, at resolution ϵ)

Theorem 5 handles the finite discrete case exactly. The continuous case — $x \subseteq \mathbb{R}^d, z \subseteq \mathbb{R}^{d_Z}, q$ continuous — cannot be resolved *exactly* (pointwise recovery of a continuous function needs infinitely many samples). The honest form of the continuous claim is "recovery *up to resolution ϵ* ", and it reduces cleanly to Theorem 5 by ϵ -covering.

Setup (Theorem 6). Let

- $z \subseteq \mathbb{R}^{d_Z}$ be bounded with diameter D_Z ;
- $q : X \rightarrow z$ be measurable, and $\mathcal{A}_\epsilon \subseteq z$ a minimal ϵ -net of z under a chosen metric; write z_ϵ for the induced partition of z into at most $N_\epsilon := |\mathcal{A}_\epsilon|$ Voronoi cells and $q_\epsilon := \pi_\epsilon \circ q : X \rightarrow z_\epsilon$ for the ϵ -quantised true partition;
- P_X be a probability on x with *fibre balance at scale ϵ* : $\min_{V \in z_\epsilon} P_X(q^{-1}(-1)(V)) \geq 1/(c N_\epsilon)$ for some $c \geq 1$;
- $\{Y_\alpha : X \rightarrow \mathcal{Y}_\alpha\}_{\alpha=1..K}$ be deterministic tasks factoring through q that *separate z at scale ϵ* : for every pair of distinct cells $V \neq V' \in z_\epsilon$, there exists α such that Y_α is constant on $q^{-1}(-1)(V)$ and on $q^{-1}(-1)(V')$ with distinct values (i.e. the lifted labellings $h_\alpha : z \rightarrow \mathcal{Y}_\alpha$ distinguish every ϵ -cell pair).

Algorithm. Empirical common-sufficient clustering, unchanged: form response profiles $\pi(x)$, partition x by profile equality, return $\hat{q}$.

Theorem 6 (Continuous-case learnability, at resolution ϵ). *Under the setup above, for any $\epsilon_{\text{rel}} \in (0, 1)$, $\hat{q}$ recovers q_ϵ exactly (i.e., $\hat{q}(x) = \hat{q}(x')$ iff $q(x)$ and $q(x')$ lie in the same ϵ -cell) with probability $\geq 1 - \epsilon_{\text{rel}}$ from*

$$N \geq c \cdot N_\epsilon \cdot \ln(N_\epsilon / \epsilon_{\text{rel}})$$

samples. For $z \subseteq \mathbb{R}^{d_Z}$ bounded, $N_\epsilon = O((D_Z / \epsilon)^{d_Z})$, giving

$$N = O(c \cdot (D_Z / \epsilon)^{d_Z} \cdot d_Z \cdot \log(D_Z / (\epsilon \cdot \epsilon_{\text{rel}}))).$$

Proof. Apply Theorem 5 to the discretised experiment $(X, \{Y_\alpha\}, q_\epsilon, P_X)$. Balance-at-scale- ϵ and separation-at-scale- ϵ are the discrete hypotheses of Theorem 5 for $M = N_\epsilon$, c as given; the $(D_Z / \epsilon)^{d_Z}$ bound on N_ϵ is the standard ϵ -covering number of a bounded set in $\mathbb{R}^{d_Z}$. $\square$

Corollary (rate). The bound is *polynomial in $1/\epsilon$* at fixed d_Z and *exponential in d_Z* at fixed ϵ . It matches (up to log factors) the information-theoretic ϵ -covering lower bound for identifying a continuous partition from i.i.d. samples: nothing better is possible in this setting without stronger assumptions on q .

Fundamental limit (Corollary). *Uniform sample complexity polynomial in d_Z alone — i.e. dropping the exponential dependence on d_Z at fixed ϵ — is impossible without additional inductive bias on q .* This is the mathematical form of the "curse of dimensionality" for common-sufficient clustering; it is not a defect of the specific algorithm but a consequence of ϵ -covering lower bounds and Locatello's (2019) impossibility for fully unsupervised disentanglement. Escaping the curse

requires structural assumptions such as linearity (linear ICA), sparsity of the mixing map (independent-mechanism analysis, Gresele et al. 2021), exponential-family conditional latents with auxiliary variables (iVAE, Khemakhem–Kingma–Monti–Hyvärinen 2020), or interventional data on z (causal representation learning, Ahuja–Mahajan–Wang–Bengio 2022). Each supplies an inductive bias sufficient to convert the exponential-in- d_z covering bound into polynomial complexity for that specific hypothesis class.

The Cross-task Learnability Continuous instrument (Instrument 6, §4.6) exhibits Theorem 6 numerically across the grid $(d_z, r) \in \{1, 2\} \times \{4, 8, 16\}$ on the standard ε -quantised $[0, 1]^2$ world, verifying the bound at every point and exhibiting the exponential-in- d_z scaling as an exact numerical fact.

2.6 SIC in the honest split

Theorems 1–6 collapse the Structural Intelligence Conjecture from a single-conjecture ambition to a fully-derived skeleton with one empirical antecedent and one residual inductive-bias question:

- **SIC-A — Existence.** Theorem 1 + Proposition 3. The master fibration exists as a mathematical object for any well-posed task. *Theorem.*
- **SIC-B — Cross-task stability.** Theorem 4 (conditional). Stability of a single q across a task family is equivalent to the family admitting a shared Markov screen. *Conditional theorem; antecedent empirical about the world / task ensemble.*
- **SIC-C-a — Learnability, discrete.** Theorem 5. Under separation and fibre balance, empirical common-sufficient clustering recovers q at $O(c_M \log(M/\varepsilon))$ samples. *Theorem, with numerically sharp constant (Instrument 5).*
- **SIC-C-b — Learnability, continuous at resolution ε .** Theorem 6. Empirical common-sufficient clustering recovers q at resolution ε from $O(c \cdot (D_z/\varepsilon)^{d_z} \log((D_z/\varepsilon)^{d_z}/\varepsilon_{\text{rel}}))$ samples. *Theorem; polynomial in $1/\varepsilon$ at fixed d_z , exponential in d_z at fixed ε (Instrument 6).*
- **SIC-C-c — Uniform polynomial in d_z .** Requires additional inductive bias (linearity, sparsity, exponential-family auxiliary, interventional data); impossible without one, per ε -covering lower bounds and Locatello (2019). *Open in general; theorem for specific hypothesis classes in the identifiable-representation-learning line — not proved from scratch in this paper.*

What remains genuinely open is only SIC-C-c — the question of which inductive biases make polynomial-in- d_z learnability possible for *which* continuous hypothesis classes. That is a mainstream question in representation-learning theory and is beyond this paper’s scope; the paper’s contribution is to identify the boundary precisely and prove learnability right up to it.

The remainder of the paper states the SIC in its full-strength form (§3), exhibits its six exact instruments (§4), sketches ten further constructs generated by the master object (§5), and closes with an honest construction target for conscious and reliable agents (§6).

3. The conjecture

Structural Intelligence Conjecture. A finite adaptive system’s central capability is to discover a quotient $q : x \rightarrow z$ such that (1) the task-relevant dynamics descend to z , (2) irrelevant variation is confined to the fibers $q^{-1}(z)$, (3) useful interventions are compactly specifiable in z , (4) those specifications re-instantiate through a compiler, and (5) their consequences remain stable across substrates and contexts.

In one line: intelligence finds the level at which the world becomes both compressible and controllable.

4. Six exact instruments

Each instrument is exact (no sampling), deterministic, and covered by a unitest suite; each is registered with a preregistration and structured provenance in the host repository. Instruments 1, 4, 5, and 6 are computational witnesses of Theorems 1, 4, 5, and 6 respectively; instruments 2 and 3 establish auxiliary dissociations on solvable cases.

4.1 Fiber Finder (`experiments/representation_search`)

Over all 2^n worlds of an n -bit Boolean space with a known ground-truth invariant, we enumerate a lattice of candidate quotients (constant, every-subset parity, identity) and three selectors. Writing $H(Y \mid q(X))$ for the residual task entropy and $\log_2 |\text{image}(q)|$ for the description length:

- `minimal_sufficient` — among quotients with $H(Y \mid q(X)) = 0$, minimize description length;
- `mdl_only` — minimize description length;
- `accuracy_only` — maximize mutual information, tie-broken toward the finest map.

Result (exact). In every task, `minimal_sufficient` recovers the exact ground-truth invariant; `mdl_only` selects the constant map (insufficient — $H(Y \mid q(X)) > 0$, the obstruction collapses); `accuracy_only` selects the identity (sufficient but uncompressed, $\log_2 |\text{image}| = n$). Sufficiency, description length, and accuracy are three different quantities, and only the sufficiency-then-compress rule finds the invariant. This is the computational witness of Theorem 1: it exhibits the minimal sufficient quotient by exhaustive enumeration and shows the failure modes of the two natural alternatives. It is also the discrete, non-interventional core of the fiber audit (§5.4) and connects to the host repository's finding that *weakness* (a fiber quantity) predicts generalization where MDL (a specification-length quantity) does not.

4.2 Structure compiler (`experiments/structure_compiler`)

An abstract automaton exhibits accumulation (a level integrates an input), a phase transition (a regime flips at an up-threshold), and hysteresis (the down-threshold is lower, so the regime remembers). From a fixed input schedule it produces a deterministic trajectory of `(level, regime)` pairs. Four compilers F_i map the trajectory into music (pitch/octave), a visual field (height/hue), text (regime-keyed lexicon, line length $\propto$ level), and spatial navigation (a corridor with regime-gated edges); each has a readback q_i .

Result (exact). For every medium $q_i \circ F_i = \text{id}$ on the trajectory (fidelity 1.0), and all media read back to the *same* abstract trajectory: the four embodiments are verifiably one work, not four mood-matched artefacts. The structure genuinely contains the motifs (a mid-level appears in both regimes). Fidelity is 1.0 *by construction* — the instrument tests that the compilers are faithful functors, and is the base case for the autocatalytic work of §5.10.

4.3 Agency benchmark (`experiments/symbolic_causation`)

An exact finite-state world with target and failure regions; futures are enumerated to the horizon. A symbolic model m is an operation on the trajectory distribution $P(Y \mid s)$. We measure **signal** $\Delta_{KL} = KL(P(Y|m) \parallel P(Y|\text{baseline}))$, **control** `goal_gain`, **knowledge** `predictive_accuracy`, and **agency** (control + small `calibration_error` between claimed and true do-effect + positive `transfer` under a perturbation the intervention did not choose).

Result (exact). Seven hand-built conditions realize distinct metric signatures recovered by a fixed classifier: a `noise_signal` condition moves the distribution with zero control; a `knowledge_only` condition predicts with zero signal; a **false_credit** condition improves the observed outcome while its true do-effect is zero and its self-attribution is miscalibrated; and a brittle controller controls but does not transfer. No single scalar of behavioural influence identifies agency. This is the measurement core for concern geometry (§5.1), causal semantics (§5.7), and ensemble-governance alignment (§5.9).

4.4 Cross-task sufficiency (`experiments/cross_task_sufficiency`)

On the 4-bit Boolean world $x = \{0, 1\}^4$ with latent $z(x) = (\text{parity}\{0, 1\}(x), \text{parity}\{2, 3\}(x))$, we enumerate a lattice of quotients that includes single-bit reads, all subset parities, joint pair-parities (including z itself), joint bit reads, and the identity. Two task families:

- *Shared through* z : `parity{0,1}`, `parity{2,3}`, `parity{0,1,2,3}` — every task factors through z .
- *Not shared beyond* z : `bit_0`, `bit_1`, `bit_2`, `bit_3` — each task reveals a single coordinate.

For each family and each quotient we compute exact per-task conditional entropies and common-sufficiency status; the coarsest common sufficient statistic (CSS) is the smallest-image quotient that is sufficient for every task in the family.

Result (exact). For the shared family, the coarsest CSS is exactly $\text{joint}(\text{parity}\{0,1\}, \text{parity}\{2,3\}) = z$ (image size 4, description length 2 bits, strictly less than $\log_2|x| = 4$ bits). For the not-shared family, the coarsest CSS is the identity map (image size 16, no compression). For the shared family, each single task's minimal sufficient statistic is a 1-bit parity (image size 2), strictly coarser than the family CSS's 4-class partition: combining tasks tightens the required partition by a factor of two.

This is the computational witness of Theorem 4. Cross-task stability — the world's tasks admitting a single shared quotient — is *not* a property the system posits; it is a property of the task family. When the family admits a common Markov screen, the CSS collapses to the screen. When it does not, the CSS is the identity. The dissociation is exact.

4.5 Cross-task learnability (experiments/cross_task_learnability)

Building on the shared task family of Instrument 4 (which is verified to *separate* the latent z , satisfying Theorem 5's identifiability antecedent), this instrument computes the exact recovery probability $P[\hat{q} = q]$ of the empirical common-sufficient clustering algorithm as a function of sample count N . Recovery reduces to a coupon-collector question on the fibre partition of x , whose distribution is given by inclusion–exclusion over the $2^M = 16$ subsets of the fibres — computable exactly, no Monte Carlo.

Two distributions on $x = \{0,1\}^4$:

- *uniform*: $p_{\min} = 1/4$, fibre-balance constant $c = 1$;
- *skewed*: fibre masses $(0.625, 0.125, 0.125, 0.125)$, $p_{\min} = 1/8$, $c = 2$.

Result (exact). At Theorem 5's sample bound $N = \lceil c \cdot M \cdot \ln(M/\varepsilon) \rceil$ with $\varepsilon = 0.05$, $M = 4$, the exact recovery probability is 0.9775 on the uniform distribution (bound $N = 18$) and 0.9756 on the skewed distribution (bound $N = 36$) — both strictly above $1 - \varepsilon = 0.95$. Below $M = 4$ samples recovery is impossible (pigeonhole), verified exactly. The recovery curve is monotone in N . The theorem bound is honest and only mildly loose: uniform recovery hits 0.95 already at $N = 14$, so the union- bound overhead is 4 samples in this regime.

This is the computational witness of Theorem 5. It certifies that the sample-complexity bound is not asymptotic hand-waving — for the shared task family and both a balanced and a doubly-unbalanced compiler, the bound is tight enough to hit the target error at the predicted N , and it degrades exactly as c predicts under compiler imbalance.

4.6 Cross-task learnability, continuous (experiments/cross_task_learnability_continuous)

Theorem 6's continuous-case bound, verified exactly across a $(d_Z, r) \in \{1, 2\} \times \{4, 8, 16\}$ grid on an ambient $x = [0, 1]^2$ quantised into a 16×16 grid. For each pair, the latent z is a coarser $r \times r$ (for $d_Z = 2$) or $r \times 1$ (for $d_Z = 1$) grid; fibres are exactly balanced by choosing $r \mid 16$. Recovery probability is computed by the $O(N \cdot M)$ DP recursion $f(n, k) = f(n-1, k) \cdot k/M + f(n-1, k-1) \cdot (M-k+1)/M$ (with $f(0, 0) = 1$, returning $f(N, M)$); this is numerically stable up to $M = 256$, where the log-domain inclusion–exclusion form suffers catastrophic cancellation.

Result (exact).

d_Z	r	M	N_bound	P(recover @ bound)
1	4	4	18	0.9775
1	8	8	41	0.9667
1	16	16	93	0.9609
2	4	16	93	0.9609
2	8	64	458	0.9538
2	16	256	2187	0.9521

All six grid points meet the $1 - \varepsilon_{\text{rel}} = 0.95$ target at Theorem 6's bound. Recovery is zero below M (pigeonhole) and

monotone in N .

Ratio witness. $N_{\text{bound}}(d_Z=2, r) / N_{\text{bound}}(d_Z=1, r) = 5.17, 11.17, 23.52$ at $r = 4, 8, 16$ — cleanly matching the $r \cdot (\log \text{ slack})$ prediction of Theorem 6. Going from $d_Z = 1$ to $d_Z = 2$ at $r = 16$ multiplies the required sample count by $\approx 23.5\times$; that is the curse of dimensionality made numerical.

This is the computational witness of Theorem 6. It certifies that empirical common-sufficient clustering saturates the ε -covering lower bound: it is optimal within the class of algorithms that make no inductive assumption on q . Escaping the exponential-in- d_Z scaling requires an inductive bias (linearity, sparsity, exponential-family conditional latents, interventional data) — SIC-C-c is the residual open problem, and the identifiable- representation-learning literature is where it is (partially) resolved.

5. The extended program

Each construct is a target generated by the master object; those marked *conjectural* are not proved here.

1. **Concern as fiber geometry.** A concern state reweights the compiler, $K_c(dx|z) \propto e^{\{\beta U_c(x,z)\}} K(dx|z)$, inducing an information geometry (Fisher metric) on concern states; concern transport between contexts is transport between kernels, and its holonomy measures path-dependence. Extends the host repository's *Geometry of Concern* and *Gauge-Fixed Transport of Concern*.
2. **Conditional rate–distortion control limit** (*conjectural*). Robust control of $Y = f(X)$ from a coarse spec $Z = q(X)$ requires $H(Y|Z) \approx 0$, or under tolerated distortion D , extra addressed bits $B_{\text{extra}} \geq R_{\{Y|Z\}}(D)$ — a generalization of the biology theorem explaining why exact micromanagement fails.
3. **Abstraction frontier.** The Pareto set trading task-sufficiency $I(Y;X|q(X))$, dynamical closure $I(Z_{\{t+1\}};X_t|Z_t, A_t)$, cost $H_0(Z)$, and control regret — an antichain, explaining why two representations can both be "right" yet incomparable. Seeded by §4.1.
4. **Fiber audit.** Vary the allegedly irrelevant degrees of freedom while holding q fixed and measure $\Delta_q(z) = \sup_{\{x, x' \in q^{-1}(z)\}} d(P(Y|do\ x), P(Y|do\ x'))$. The operational form of Wigner's relevant/irrelevant split and the ontology's truth-as-invariance criterion; seeded (non-interventional core) by §4.1. *Cross-task version:* Instrument 4 (§4.4) — the coarsest common sufficient statistic is the operational fiber audit at the family level.
5. **Theory atlas.** Treat theories as charts M_i on contexts U_i with translations T_{ij} ; test the cocycle $T_{jk} \circ T_{ij} = T_{ik}$. Where it fails, the obstruction is informative — a sheaf/stack view of theory integration for Wigner's non-unification worry.
6. **Compiler tomography.** Infer the shared compiler and compact specs from many (s_i, x_i) by MDL; then compiler ecology $K_{\{t+1\}} = U(K_t, \text{outcomes})$ — build a compiler under which good outcomes are cheap, the formal language for education, institutions, and long-horizon training.
7. **Causal semantics.** Two symbols are equivalent when they induce naturally equivalent update operators $\Psi_{\{m,c\}}$ across independent contexts — a meaning layer that ordinary co-occurrence embeddings omit. Extends §4.3.
8. **Representation-repair calculus.** A library mapping failure signatures to minimal structural lifts (scalar→operator, global norm→localized measure, quotient→restored fiber, static→path space, affine→projective, point→ensemble, non-composing→interface, symmetry→gauge-fix). Reads the machine discovery notes directly.
9. **Alignment as ensemble governance** (*conjectural*). Target a viable region $V \subseteq Z$ with $\Pr[q(X_t) \in V \ \forall t] \geq 1-\delta$ under a broad family of unresolved compiler/environment states; a fiber audit is the alignment evaluation.
10. **Autocatalytic artwork.** $S_t \rightarrow K_t \xrightarrow{E_t \rightarrow (\text{experience})} K_{\{t+1\}}$: early movements teach the grammar by which later movements become legible. Extends §4.2.

6. Capstone: conscious and reliable agents, honestly

The program bears on two natural questions — can agents be made conscious, and can they be made never wrong — and the disciplined answer to both is *not in the literal sense*. The strongest defensible construction factors as two nested systems.

Concerned Self-Modeling Core — an active, self-maintaining fixed point of the coarse-graining loop: a persistent agent

`tests.test_cross_task_learnability_continuous`

Full development, the stochastic-fibration formalism, and the ten constructs are in `notes/structural_intelligence_conjecture.md`; the underlying six-work synthesis is in `notes/latent_structures_meta_framework.md`. An interactive demo of the six instruments is served from `sites/structural_observatory`.